IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM BOB and IBM Granite Models”

Project Tile – [AI Health Symptom Assessment Agent]

Domain of Project – [Healthcare]

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Problem Statement -

AI Health Symptom Assessment Agent

The Challenge

Many individuals struggle to understand the significance of their symptoms and whether they require medical attention.

The Objective

Develop an AI-powered Health Symptom Assessment Agent that interacts with users, analyzes reported symptoms, identifies possible health conditions, assesses risk levels, and provides appropriate healthcare recommendations. The solution should:

- **Analyze Symptoms** – Collect and analyze user-reported symptoms to understand their health condition.
- **Identify Possible Conditions** Predict possible health conditions based on the user's symptoms and clinical information.
- **Assess Risk Levels** – Evaluate symptom severity and classify the patient's condition as Low, Moderate, or High Risk.
- **Provide Recommendations** – Generate personalized healthcare guidance and recommend appropriate next steps based on the assessment.

Proposed Solution -

Proposed Solution: AI Health Symptom Assessment Agent

The proposed solution is an **AI-powered Health Symptom Assessment Agent** built using IBM BOB, IBM watsonx.ai, and the IBM Granite 3 H Small model to provide intelligent, real-time health assessments. The system interacts with users, analyzes their symptoms, evaluates risk levels, and provides personalized healthcare recommendations.

The solution uses Large Language Model (LLM) capabilities to understand natural language symptom descriptions, interpret medical history, and generate clinically relevant responses. The AI-powered assessment helps users better understand the seriousness of their symptoms while encouraging timely medical consultation.

The system consists of four key agents:

- **Symptom Analysis Agent** collects patient information such as age, gender, symptoms, duration, medical history, and current medications, then analyzes the reported symptoms.
- **Condition Identification Agent** compares the analyzed symptoms with medical knowledge to identify possible health conditions and presents the most relevant possibilities without making a definitive diagnosis.
- **Risk Assessment Agent** evaluates symptom severity, patient history, and identified conditions to classify the health risk as **Low, Moderate, or High**, while detecting emergency warning signs.
- **Recommendation Agent** generates personalized healthcare recommendations, including home-care advice, medical consultation, lifestyle suggestions, and emergency guidance based on the assessed risk level.

The solution provides a user-friendly web dashboard that displays patient information, clinical summaries, possible conditions, risk scores, red-flag symptoms, and healthcare recommendations in an easy-to-understand format.

IBM BOB is used to orchestrate the AI workflow, while the IBM Granite 3 H Small model performs intelligent symptom analysis and recommendation generation through natural language understanding.

Overall, the proposed solution delivers an intelligent healthcare assistant that promotes early symptom assessment, informed healthcare decisions, and timely medical intervention, helping users access reliable preliminary health guidance anytime and anywhere.

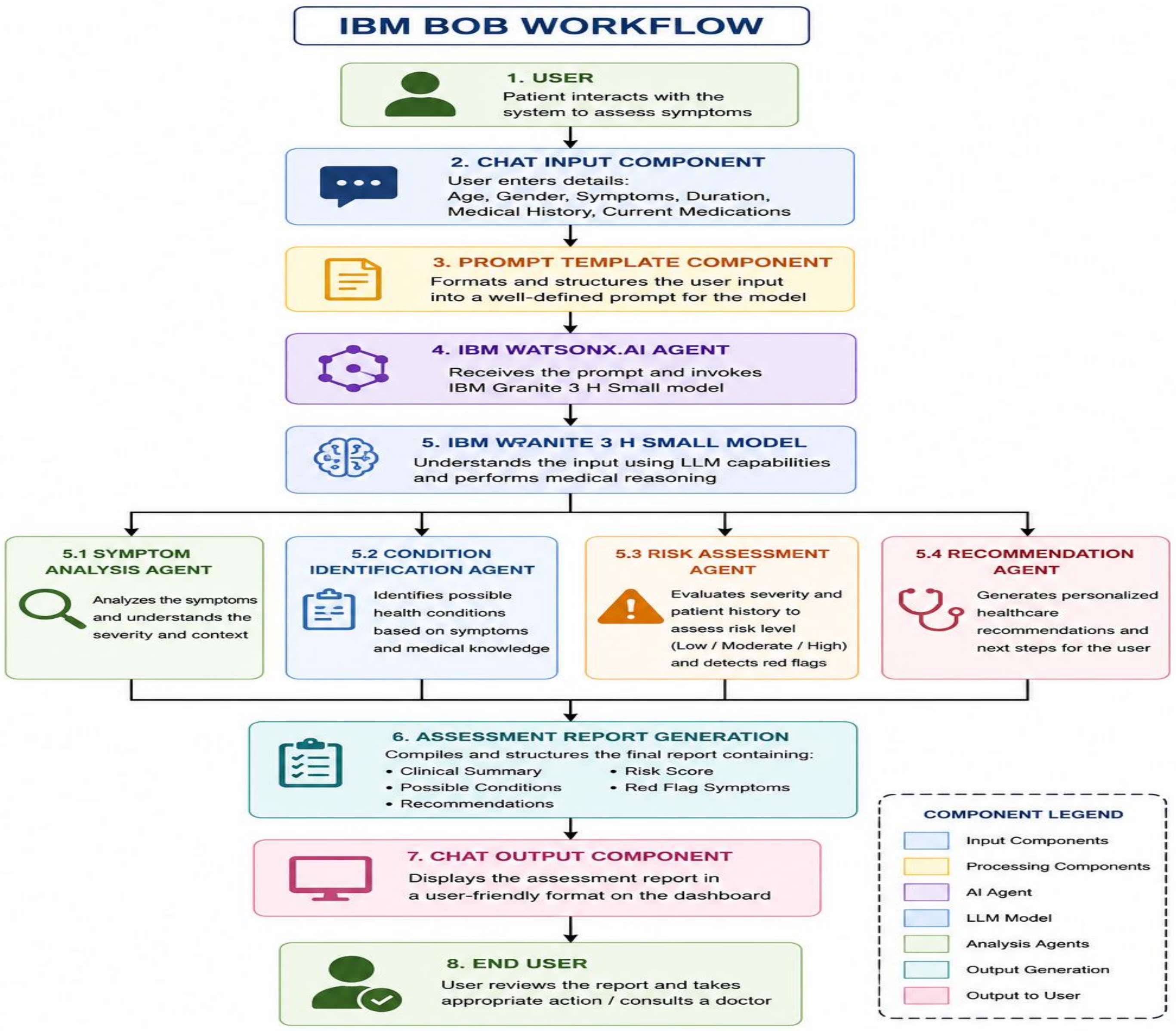
Technology Used

- **IBM BOB Platform** – Used to design, develop, and orchestrate the AI Health Symptom Assessment Agent workflow, enabling intelligent interaction between different AI components.
- **IBM Granite 3 H Small Model** – Provides natural language understanding, symptom analysis, clinical reasoning, and generation of personalized healthcare recommendations based on user inputs.
- **IBM Cloud** – Offers a secure, scalable, and reliable cloud infrastructure for deploying, hosting, and managing the AI Health Symptom Assessment Agent.
- **IBM watsonx.ai** – Used for accessing and deploying the IBM Granite model, enabling prompt execution, model inference, and AI-powered healthcare assessments.
- **Prompt Engineering** – Carefully designed prompts guide the Granite model to accurately analyze symptoms, assess risk levels, identify possible conditions, and generate appropriate recommendations.
- **HTML, CSS & JavaScript** – Used to build a responsive and user-friendly web interface for collecting patient information and displaying assessment reports.
- **REST API Integration** – Connects the frontend application with IBM watsonx.ai services, enabling seamless communication between the user interface and the AI model.

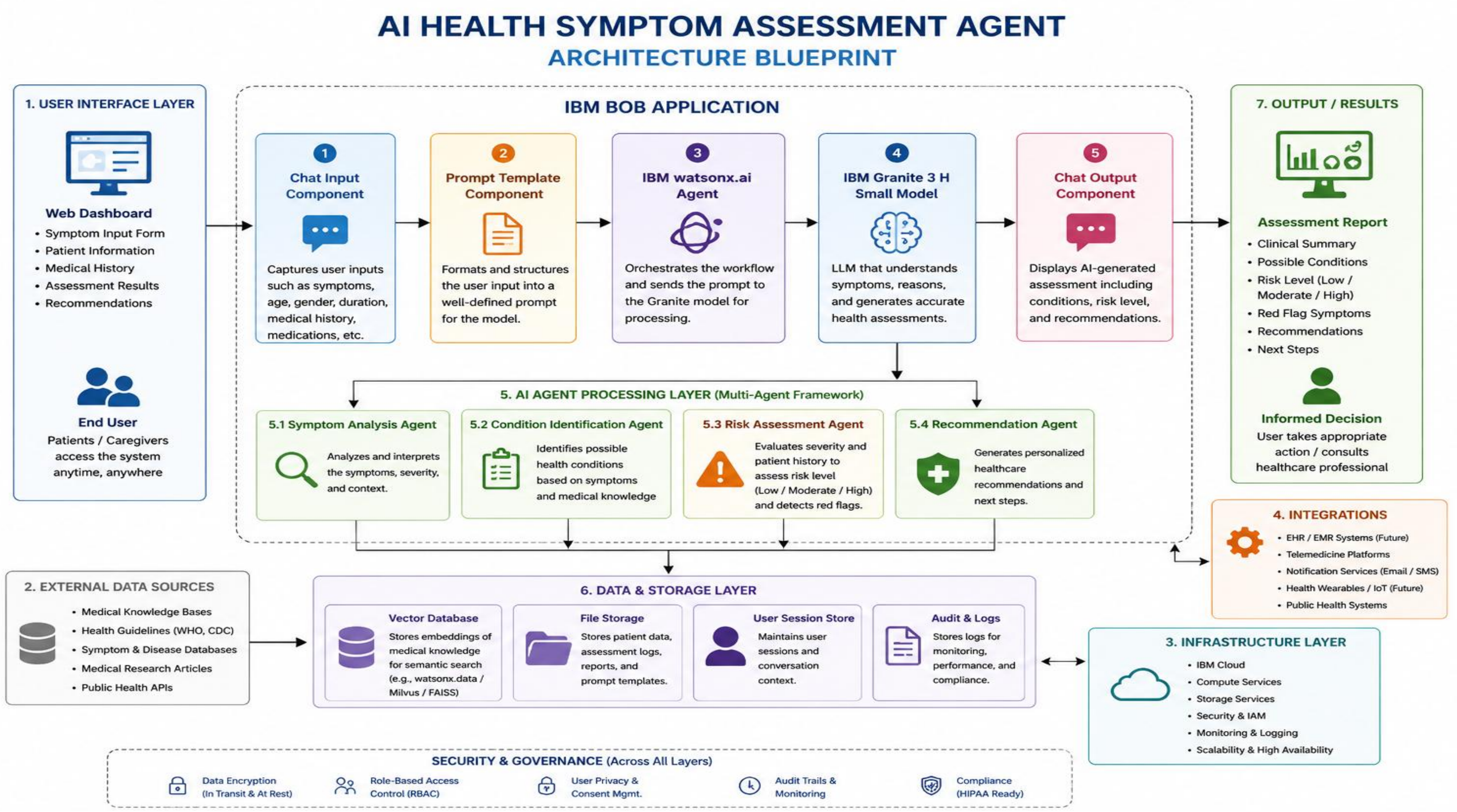
IBM BOB component Used -

1. **Chat Input** – Provides an interactive interface where users enter patient details such as age, gender, symptoms, symptom duration, medical history, and current medications for health assessment.
2. **IBM watsonx.ai Agent** – Processes the user input using the IBM Granite 3 H Small model to analyze symptoms, identify possible health conditions, assess risk levels, and generate healthcare recommendations.
3. **IBM Granite 3 H Small Model** – Performs natural language understanding, clinical reasoning, symptom interpretation, and generates accurate, context-aware healthcare responses in real time.
4. **Chat Output** – Displays the AI-generated assessment report, including the clinical summary, possible medical conditions, risk score, red-flag symptoms, and personalized healthcare recommendations.
5. **File Component** – Stores and retrieves patient information, symptom records, assessment reports, and prompt templates required for generating consistent and reliable AI responses.
6. **Prompt Template Component** – Uses carefully designed prompts to guide the Granite model in collecting symptom information, performing clinical analysis, classifying risk levels, and generating structured healthcare recommendations.

IBM BOB workflow -



Architecture blueprint



Role of Agentic AI in AI Health Symptom Assessment Agent

01 **Agentic AI** enables the AI Health Symptom Assessment Agent to function as an intelligent, autonomous healthcare assistant rather than a simple symptom checker. Using **IBM watsonx.ai** and the **IBM Granite 3 H Small** model, the system understands natural language inputs, analyzes patient symptoms, and coordinates multiple AI-driven tasks to provide meaningful healthcare guidance.

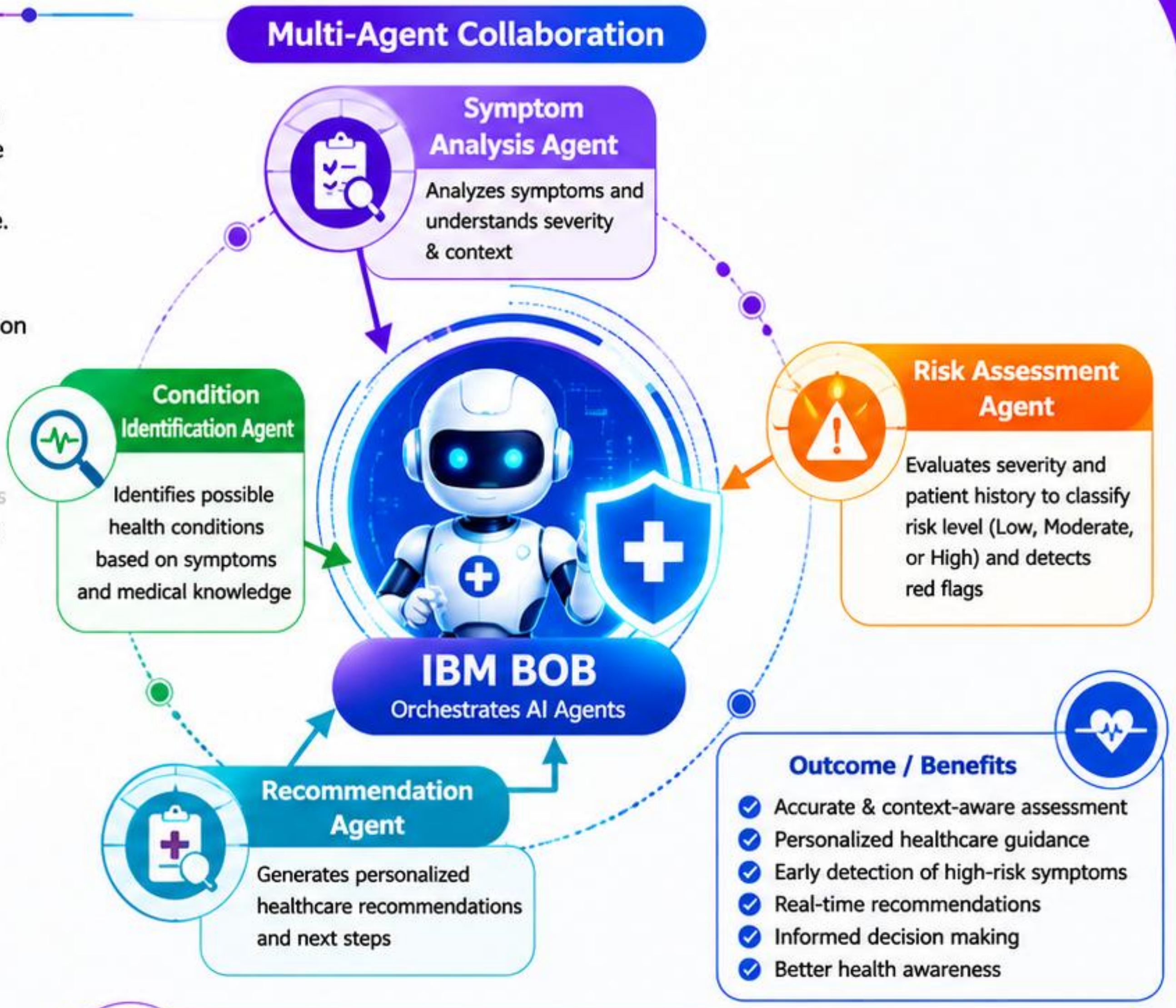
02 The solution employs **specialized AI agents** that collaboratively perform symptom analysis, condition identification, risk assessment, and recommendation generation. Each agent focuses on a specific healthcare task while sharing information with the others, ensuring accurate, context-aware, and reliable assessments.

03 Agentic AI maintains **context awareness** by considering patient details such as age, gender, symptom duration, medical history, and current medications before generating responses. This enables the system to deliver personalized health assessments instead of generic advice.

04 The AI continuously performs **reasoning and decision-making** to evaluate symptom severity, identify possible health conditions, classify the patient's risk level (**Low**, **Moderate**, or **High**), and detect red-flag symptoms requiring immediate medical attention.

05 **IBM BOB** orchestrates the interaction between the AI agents, while **IBM Granite 3 H Small** provides advanced language understanding and medical reasoning to generate structured clinical summaries and healthcare recommendations in real time.

06 Overall, Agentic AI transforms the application into an **intelligent digital health assistant** that supports early symptom assessment, improves healthcare awareness, assists users in making informed decisions, and encourages timely consultation with qualified medical professionals.



Agentic AI transforms the system into an

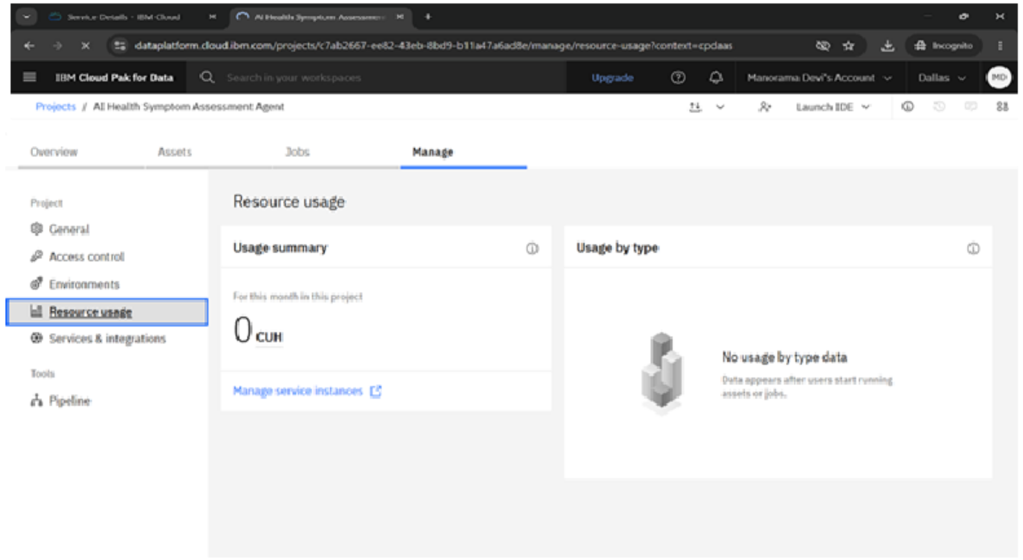
Intelligent Digital Health Assistant

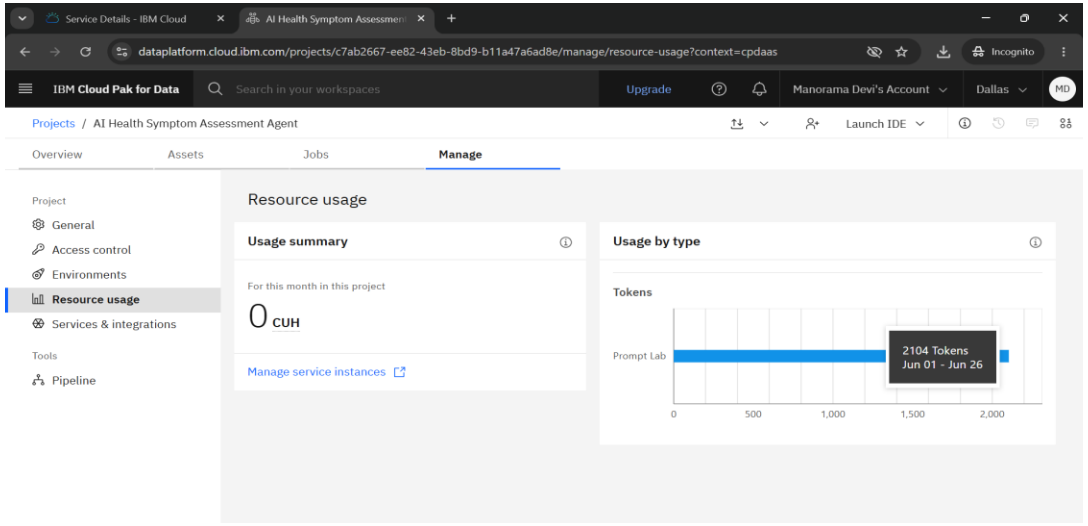
Token Usage

2104 token are used

Before Prompt

After the Prompt





Novelty and Uniqueness

- The proposed solution leverages **Agentic AI** to provide an intelligent, context-aware health assessment rather than a traditional symptom checker.
- It integrates **IBM BOB**, **IBM watsonx.ai**, and the **IBM Granite 3 H Small** model for autonomous symptom analysis and healthcare guidance.
- The system considers **patient age, gender, symptom duration, medical history, and current medications** to generate personalized assessments.
- It performs **multi-step AI reasoning**, including symptom analysis, possible condition identification, risk assessment, and recommendation generation.
- The application classifies health conditions into **Low, Moderate, and High Risk**, helping users understand the urgency of their symptoms.
- It provides **real-time, personalized healthcare recommendations** instead of generic advice.
- The interactive dashboard presents **clinical summaries, possible conditions, red-flag symptoms, and recommendations** in a user-friendly format.
- The modular architecture enables future integration with **electronic health records (EHRs), telemedicine platforms, wearable devices, and other healthcare systems**, making the solution scalable and extensible.

Git Hub Link

https://github.com/Manorama-Devi/AI-Health-Symptom-Assessment-Agent

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Future Scope

- Voice input
- RAG integration
- Appointment booking
- Multi-agent collaboration

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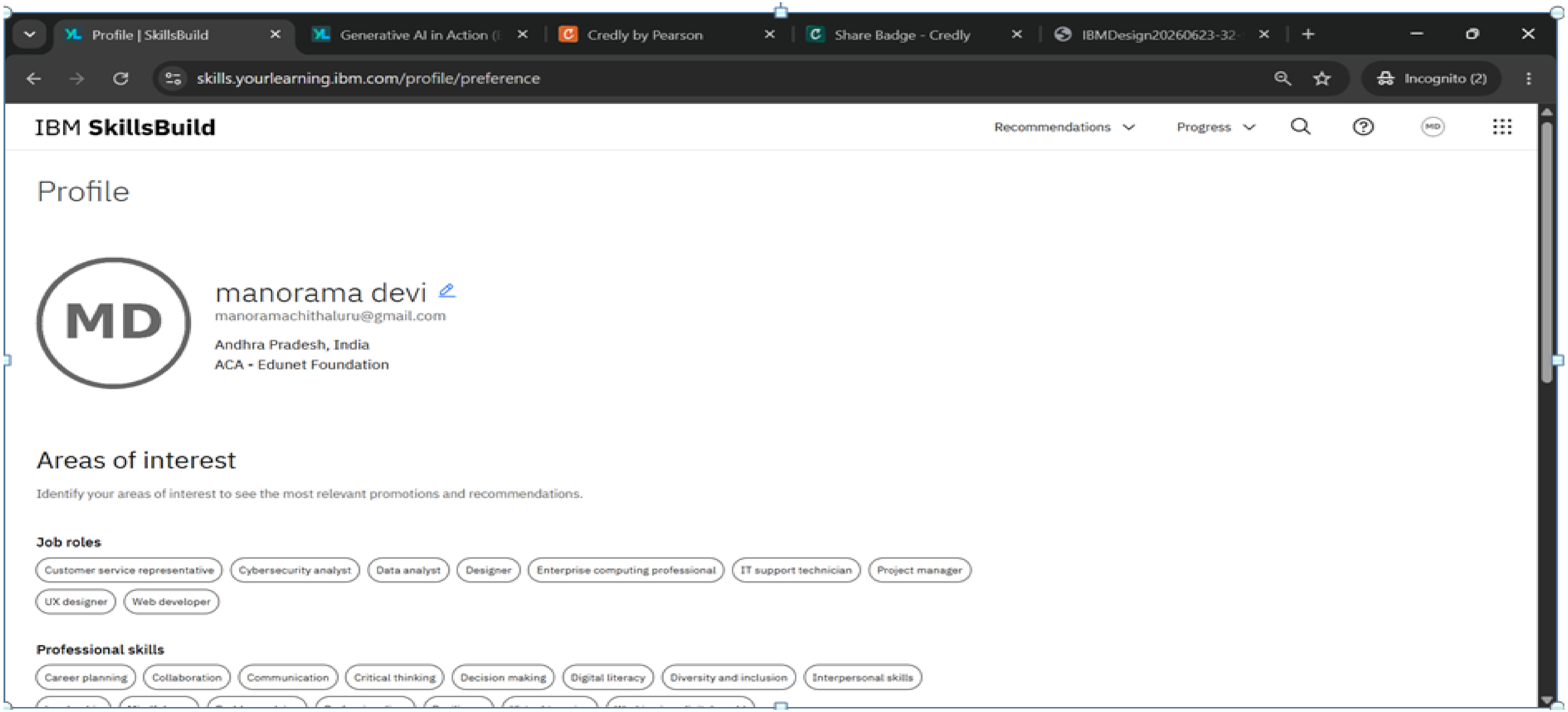

Thank You!

Thank you for your time and interest.